

MongoDB Engineer Field Manual & Runbook

Troubleshooting Guidelines, Diagnostic Workflows & Chaos Drills

AUDIENCE TARGET: DATABASE & INFRASTRUCTURE ENGINEERS

1. Cluster Topology & Quick Port Reference

This field manual contains operational runbooks, diagnostic commands, and local chaos drill instructions for supporting our 14-node MongoDB sharded cluster.

Service / Component	Replica Set / ID	Default Port	Role / Notes
mongos Router 1	mongos01	27017	Primary Query Router
mongos Router 2	mongos02	27027	Secondary Query Router (HA)
Config Server Set	csrs (csrs01-03)	27019	Metadata Storage
Shard 01 Data Set	shard01 (s1-n1..3)	27018	Data Shard 1
Shard 02 Data Set	shard02 (s2-n1..3)	27018	Data Shard 2
Shard 03 Data Set	shard03 (s3-n1..3)	27018	Data Shard 3

2. Failure Diagnosis & Emergency Repair Runbooks

Scenario 2.1: Primary Node Down

Run `rs.status()` on remaining secondaries to confirm state. If election stalls, verify majority network connectivity.

```
rs.status();
db.printReplicationInfo();
```

Scenario 2.2: Majority CSRS Quorum Loss

If 2 CSRS nodes fail, metadata turns read-only. Force reconfig remaining node:

```
cfg = rs.conf();
cfg.members = [{ _id: 0, host: 'csrs01:27019', priority: 1, votes: 1 }];
rs.reconfig(cfg, { force: true });
```

Scenario 2.3: Oplog Rollover / Reporting Lag

Dynamically expand oplog size and enforce retention hours on live nodes:

```
db.adminCommand({ replSetResizeOplog: 1, size: 102400, minRetentionHours: 48.0 });
```

3. Local Sandbox & Chaos Engineering Drills

Run the local chaos generator (`mongodb_chaos_generator.py`) to practice failure drills:

```
# Inject Shard Primary Crash
python3 mongodb_chaos_generator.py --scenario 1
```

```
# Inject CSRS Majority Failure  
python3 mongodb_chaos_generator.py --scenario 2
```

```
# Restore Cluster  
python3 mongodb_chaos_generator.py --heal
```

After completing a drill, document your RCA in the ticketing system using `rca_template.md`.